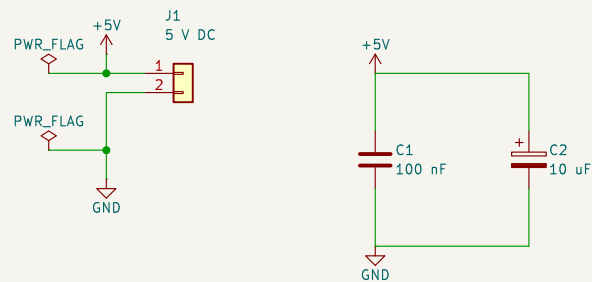


NE555 / LED BLINKER

5 V DC | approx. 1 blink / second | through-hole parts

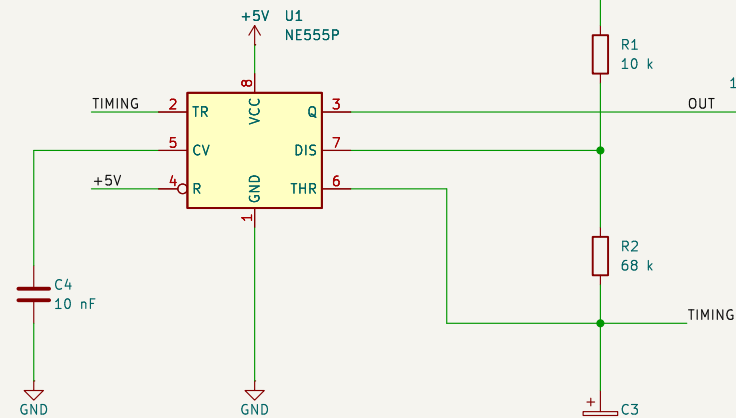
01 POWER & BYPASS



C1: place close to U1 pins 8 / 1.

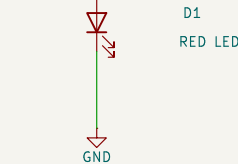
Use a regulated, isolated 5 V source.
J1 pin 1 = +5 V; pin 2 = GND.
Observe C2 / C3 and LED polarity.

02 ASTABLE TIMER



RESET held high.
C4 filters control voltage.

03 INDICATOR



LED is ON when OUT is HIGH.

TIMING | calculated nominal values

$T = 0.693 \times (R1 + 2 \times R2) \times C3 = 1.012 \text{ s}$ $f = 0.988 \text{ Hz}$
LED on: 0.541 s / off: 0.471 s / duty: 53.4%

Timing varies with capacitor tolerance. Not physically tested.
Reference: Texas Instruments NE555 datasheet, astable operation.

Electrical rules checked; not physically built or tested.
Nominal 0.99 Hz | Low-voltage educational circuit
dot | KiCad demonstration

Sheet: /
File: NE555_LED_Blinker.kicad_sch

Title: NE555 – LED blinker / 5 V

Size: A4 Date: 2026-10-09

KiCad E.D.A. 9.0.2

Rev: 1.0

Id: 1/1